



Modelling Financial Markets

Chapter 19: Review and Project Presentations



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Acronyms used in this material (1/2)

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ ASE: Academia de Studii Economice din București (the Bucharest University of Economic Studies)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CI: Confidence Interval
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ CV: Cross-Validation
- ▣ DCC: Dynamic Conditional Correlation
- ▣ ES: Expected Shortfall
- ▣ ETF: Exchange-Traded Fund
- ▣ EVT: Extreme Value Theory
- ▣ FDR: False Discovery Rate
- ▣ FHS: Filtered Historical Simulation
- ▣ FRED: Federal Reserve Economic Data
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ FWER: Family-Wise Error Rate
- ▣ FX: Foreign eXchange
- ▣ FZ0: Fissler–Ziegel loss, zero-homogeneous version
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-EVT: GARCH filter combined with Extreme Value Theory for the tail
- ▣ GRS: Gibbons–Ross–Shanken (test)

Acronyms used in this material (2/2)

- ▣ GSADF: Generalised Sup Augmented Dickey–Fuller (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HS: Historical Simulation
- ▣ ML: Machine Learning
- ▣ MSE: Mean Squared Error
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ SML: Security Market Line
- ▣ SRISK: Systemic RISK (capital shortfall measure)
- ▣ US: United States
- ▣ VIX: CBOE Volatility Index

Outline

Introduction

Review I: Markets, Returns and Pricing

Review II: Volatility, Dependence and Risk

Review III: High Frequency, Continuous Time and Options

Review IV: Learning, Text and New Markets

Common Pitfalls

Integrated Case Study: BET

The Team Project

Using AI in Your Project

Project Seeds

AI for Scientific Discovery

Conclusions

References

What This Chapter Does

Purpose

- Review Chapters 0–18 as one connected story
- Collect the pitfalls that returned in every chapter
- Run one integrated case study, from raw prices to a backtested value-at-risk (VaR) at 1%
- Prepare the team project: its four steps, the question, the methods and the AI policy

Assessment

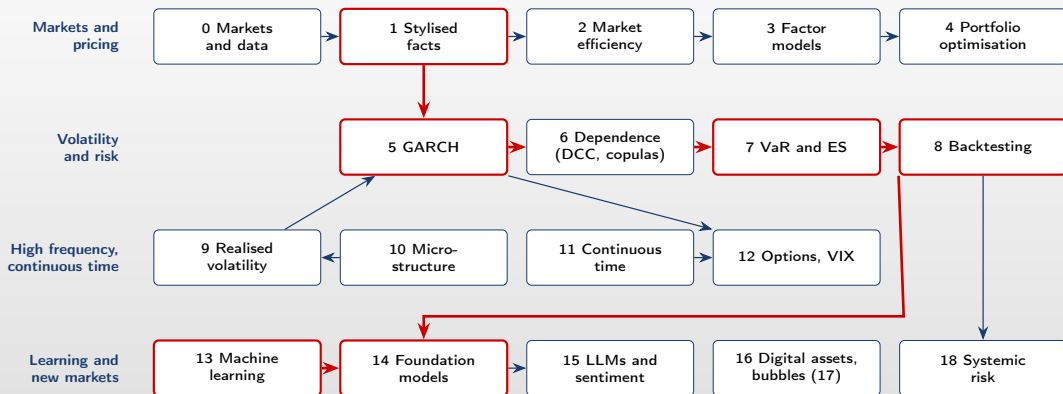
- Team project (GitHub, presentation, oral defence): **70%**
- Quizzes and seminar activity: **20%**
- Attendance: **10%**; there is no final exam



Bucharest University of Economic Studies (ASE), Piața Romană

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The Course in One Picture



- Red: the main thread of the course, from stylised facts to risk forecasts that we can test
- Blue arrows: where one chapter supplies the input or the benchmark of another

Reading the Course Map

- Chapter 1 sets the targets: heavy tails, volatility clustering, almost no linear predictability
 - ▶ Every later model is judged by whether it reproduces these facts
- Chapters 5–8 turn the facts into a risk number and a test of that number
 - ▶ GARCH gives σ_{t+1} ; VaR and ES give the loss; backtesting says whether to trust it
- Chapters 9–12 measure volatility better (intraday data, option prices) and in continuous time
- Chapters 13–15 bring flexible models, always evaluated with the tools of Chapter 8
- Chapters 16–18 apply the same toolkit to crypto, bubbles and the whole financial system

Five Questions for Every Model

Before estimating

- **Data:** which series, which calendar, which adjustments (dividends, splits, holidays)?
- **Model:** which stylised fact does it capture, and which does it ignore?
- **Estimation:** on which window, with which information at time t ?

After estimating

- **Evaluation:** out of sample, with a loss that matches the target (QLIKE, FZ0, Kupiec)?
- **Uncertainty:** robust standard errors, bootstrap, a test against a simple benchmark?
- If one answer is missing, the result is not yet a finding

Chapter 0: Financial Markets in 2026

Key idea

- Markets combine exchanges, over-the-counter dealers, exchange-traded funds (ETFs), algorithms and blockchains
- Data quality is a modelling decision: source, adjustment and calendar change the result

Key formula

- $\text{CAGR} = (P_T/P_0)^{1/Y} - 1$ (compound annual growth rate)
- $\sigma_{\text{ann}} = \sqrt{P} \hat{\sigma}$

Result from the data

- 2015–2026: SPY (S&P 500 ETF, dividends reinvested) 13.7% a year with 17.6% volatility; Bitcoin 60.6% a year with 66.7%
- BET, July 2007 to February 2009: a drawdown of −82.5%

Leads to

- Returns, not prices, are the object of study (Chapter 1)

Chapter 1: Stylised Facts and Returns

Key idea

- ▣ Returns are uncorrelated but not independent: their size is predictable, their sign almost not
- ▣ Heavy tails and volatility clustering in every market
Cont (2001)

Key formula

- ▣ $\hat{K} - 3 = \frac{T^{-1} \sum_t (r_t - \bar{r})^4}{\hat{\sigma}^4} - 3$
- ▣ $Q(m) = T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k}$

Result from the data

- ▣ Excess kurtosis from 9.3 (gold) to 17.8 (BET-TR); the Normal distribution gives 0
- ▣ Days beyond 5σ are more than 1000 times more frequent than the Normal distribution predicts

Leads to

- ▣ Volatility models (Chapter 5) and tail risk measures (Chapter 7)

Chapter 2: Market Efficiency

Key idea

- Efficiency: excess returns are unpredictable given the information set [Fama \(1970\)](#)
- Efficiency varies over time (adaptive markets); tests need robust inference

Key formula

- $$VR(q) = \frac{\text{Var}(r_t + \dots + r_{t-q+1})}{q \text{Var}(r_t)} = 1$$
 (variance ratio) under a random walk

Result from the data

- Share of rolling windows that reject the random walk: S&P 500 6.4%, BET 22.4%, Bitcoin 0.0%
- Statistical rejection is not exploitable profit

Leads to

- Anomalies and factors (Chapter 3); the multiple-testing problem

Chapter 3: Factor Models and Asset Pricing

Key idea

- ▣ Expected returns should reward only exposure to systematic factors
- ▣ Hundreds of published factors require higher hurdles Harvey, Liu & Zhu (2016)

Key formula

- ▣ $E[R_i] - r_f = \beta_i (E[R_m] - r_f)$
- ▣ $\beta_i = \text{Cov}(R_i, R_m) / \text{Var}(R_m)$

Result from the data

- ▣ 25 size and book-to-market portfolios: SML slope -4.4% , $\text{GRS} = 4.18$: the CAPM is rejected
- ▣ Sector ETFs: the first principal component (the market) explains 66% of the variance

Leads to

- ▣ Portfolio construction (Chapter 4); factor models inside machine learning (Chapter 13)

Chapter 4: Portfolio Optimisation

Key idea

- ▣ Markowitz asks the right question, but estimation error dominates the answer [Markowitz \(1952\)](#)
- ▣ Shrinkage, constraints and risk-based rules stabilise the weights; 1/N is a hard benchmark [DeMiguel, Garlappi & Uppal \(2009\)](#)

Key formula

- ▣ $w_{\text{tan}} \propto \Sigma^{-1}(\mu - r_f \mathbf{1})$
- ▣ $SR = (\mu - r_f)/\sigma$ (Sharpe ratio)

Result from the data

- ▣ 60-month samples: median in-sample Sharpe 1.57, median true Sharpe 0.30
- ▣ 96% of the estimated tangency portfolios are truly worse than 1/N

Leads to

- ▣ Risk budgets need volatility and dependence forecasts (Chapters 5–6)

Recap: Markets, Returns and Pricing

- ▣ Clean data first: adjusted prices for ETFs and stocks, own calendar for each series, join on prices before returns
- ▣ Stylised facts are the benchmark for every model that follows
- ▣ Efficiency and factor tests need robust errors and multiple-testing corrections
- ▣ Optimisation amplifies estimation error: evaluate out of sample, against $1/N$

Chapter 5: Conditional Volatility: GARCH

Key idea

- ▣ Volatility clusters, so today's shock predicts tomorrow's variance Engle (1982), Bollerslev (1986)
- ▣ Persistence close to 1; asymmetry in equities; heavy-tailed innovations

Key formula

- ▣ $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$
- ▣ $h_{1/2} = \ln 0.5 / \ln(\alpha + \beta)$

Result from the data

- ▣ S&P 500, 2000–2026: $\alpha + \beta = 0.9947$, a half-life of 131 days
- ▣ GARCH volatility and the VIX: correlation 0.89

Leads to

- ▣ $\hat{\sigma}_{t+1}$ is the input of filtered VaR and ES (Chapter 7)

Chapter 6: Multivariate Volatility and Dependence

Key idea

- ▣ Correlations move with the macro regime; diversification is not guaranteed
- ▣ Copulas separate the margins from dependence; tail dependence drives joint losses

Key formula

- ▣ $Q_t = (1 - a - b) \bar{Q} + a z_{t-1} z_{t-1}^\top + b Q_{t-1}$
- ▣ $R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}$

Result from the data

- ▣ SPY-TLT (long US Treasury bond ETF): ρ from -0.40 to 0.11 after 2022, so a 60/40 portfolio went from 10.4% to 13.2% volatility
- ▣ Bitcoin-SPY DCC correlation: 0.04 before 2020, 0.31 since

Leads to

- ▣ Portfolio VaR and Euler contributions (Chapter 7); systemic risk (Chapter 18)

Chapter 7: Value-at-Risk and Expected Shortfall

Key idea

- ▣ VaR is a quantile of the loss, ES the average loss beyond it; ES is coherent
- ▣ Filtered historical simulation and conditional EVT react to volatility in days McNeil & Frey (2000)

Key formula

- ▣ $\text{VaR}_\alpha(X) = -q_\alpha(X)$
- ▣ $\text{ES}_\alpha(X) = -\frac{1}{\alpha} \int_0^\alpha q_u(X) du$

Result from the data

- ▣ S&P 500, daily: historical $\text{VaR}_{1\%} = 3.45\%$, $\text{ES}_{1\%} = 5.01\%$
- ▣ The Normal distribution gives only 2.80%: heavy tails matter

Leads to

- ▣ A risk forecast is useful only if it survives a backtest (Chapter 8)

Chapter 8: Backtesting and Evaluating Risk Forecasts

Key idea

- ▣ Test the number and the timing of breaches: coverage and independence [Kupiec \(1995\)](#), [Christoffersen \(1998\)](#)
- ▣ Rank (VaR, ES) forecasts with a consistent score (FZ0) and Diebold–Mariano tests [Fissler & Ziegel \(2016\)](#)

Key formula

- ▣ $LR_{uc} = -2 \ln \frac{(1 - \alpha)^{T-x} \alpha^x}{(1 - \hat{\pi})^{T-x} \hat{\pi}^x} \sim \chi_1^2$
- ▣ $\hat{\pi} = x/T$

Result from the data

- ▣ S&P 500, VaR 1%, 2007–2026: Normal 2.70%, HS 1.67%, FHS 1.23%, GARCH-EVT 1.13% breaches
- ▣ A GARCH filter plus a flexible tail wins in all four markets

Leads to

- ▣ The same tests judge realised-volatility and machine-learning forecasts (Chapters 9, 13, 14)

Recap: Volatility, Dependence and Risk

- ▣ GARCH turns volatility clustering into one-day-ahead forecasts
- ▣ Dependence is dynamic and strongest in the tails; crisis correlations need a volatility adjustment
- ▣ VaR and ES must combine a volatility filter with a heavy-tailed distribution
- ▣ A risk model is accepted only after tests of coverage, independence and ES

Chapter 9: Realised Volatility

Key idea

- ▣ Intraday prices measure daily variance (RV , realised variance); the log-HAR (heterogeneous autoregressive) model captures its long memory [Corsi \(2009\)](#)
- ▣ Returns divided by realised volatility are close to the Normal distribution

Key formula

- ▣ $RV_t = \sum_{i=1}^M r_{t,i}^2$
- ▣ $\ln RV_{t+1} = \beta_0 + \beta_d \ln RV_t + \beta_w \ln RV_t^{(w)} + \beta_m \ln RV_t^{(m)} + \varepsilon_{t+1}$

Result from the data

- ▣ SPY: kurtosis 15.73 for open-to-close returns, 2.65 after dividing by $\sqrt{RV_t}$
- ▣ log-HAR beats GARCH(1,1)- t in QLIKE by 10% for SPY ($p = 0.021$) and 22% for Bitcoin

Leads to

- ▣ Better volatility inputs for VaR; microstructure explains the noise (Chapter 10)

Chapter 10: Market Microstructure

Key idea

- ▣ Prices form in limit order books; the spread pays for processing, inventory and adverse selection
- ▣ Liquidity follows the clock and the VIX, and disappears in stress

Key formula

- ▣ $ILLIQ = \frac{1}{D} \sum_{d=1}^D \frac{|r_d|}{DVOL_d}$ (Amihud illiquidity; $DVOL_d$: traded value on day d)
- ▣ $s_{Roll} = 2\sqrt{-Cov(\Delta p_t, \Delta p_{t-1})}$ (Roll spread)

Result from the data

- ▣ Amihud ratio (bp per USD 1 million): SPY 0.0016, Banca Transilvania 24.7, Transelectrica 1024.0
- ▣ SPY: the opening 5-minute bar moves 2.06 times as much as a midday bar

Leads to

- ▣ Liquidity risk in VaR and in the project data choices

Chapter 11: Continuous-Time Models

Key idea

- Brownian motion is the limit of random walks; Itô's lemma solves the price equations
- Jumps explain the tails, stochastic volatility the clustering

Key formula

- $dS_t = \mu S_t dt + \sigma S_t dW_t \Rightarrow S_T = S_0 \exp\left((\mu - \frac{\sigma^2}{2})T + \sigma W_T\right)$

Result from the data

- S&P 500 under geometric Brownian motion: excess kurtosis 10.9 instead of 0
- 62 days beyond ± 4 standard deviations, against 0.59 expected

Leads to

- Option pricing needs only σ , not μ (Chapter 12)

Chapter 12: Options and the Volatility Surface

Key idea

- ▣ Black–Scholes prices options by hedging; implied volatility is the market's view of σ **Black & Scholes (1973)**
- ▣ Smile and skew encode the fear of crashes

Key formula

- ▣ $C = S N(d_1) - Ke^{-rT} N(d_2)$
- ▣ $VRP_t = VIX_t^2 - RV_{t,t+21}$

Result from the data

- ▣ 1990–2026: mean VIX 19.4, mean realised volatility over the next month 15.3
- ▣ The variance risk premium is positive on 86% of days, but not in crashes

Leads to

- ▣ Implied volatility as a feature for machine learning and as a stress indicator (Chapters 13, 18)

Recap: High Frequency, Continuous Time and Options

- ▣ Intraday data measure volatility; log-HAR forecasts it better than daily GARCH
- ▣ Liquidity is a risk of its own and varies by orders of magnitude across markets
- ▣ Continuous-time models are tractable, but realistic ones need jumps and random volatility
- ▣ Option prices carry forward-looking information, including a variance risk premium

Chapter 13: Machine Learning in Finance

Key idea

- Flexible models, but little signal and much noise; the finance-specific part is the process López de Prado (2018)
- Purged cross-validation, meta-labelling and deflated Sharpe ratios Bailey & López de Prado (2014)

Key formula

- $(1 - B)^d x_t = \sum_{k \geq 0} w_k x_{t-k}$
- $w_0 = 1, w_k = -w_{k-1} \frac{d - k + 1}{k}$

Result from the data

- Shuffled K-fold reports 69.4% accuracy on pure noise: leakage
- S&P 500: fractional differencing is stationary at $d^* = 0.25$; no model beats “always up”

Leads to

- Deep learning and foundation models, judged the same way (Chapter 14)

Chapter 14: Deep Learning and Time-Series Foundation Models

Key idea

- ▣ LSTMs (long short-term memory networks) and Transformers learn from sequences; foundation models forecast zero-shot
- ▣ Evaluate them with Chapter 8 tools, after the release of their weights

Key formula

- ▣ $L(RV, h) = \frac{RV}{h} - \ln \frac{RV}{h} - 1$ (QLIKE loss)

Result from the data

- ▣ SPY realised volatility, QLIKE: Chronos-2 0.231, log-HAR 0.234, GARCH- t 0.283; the first two are not significantly different
- ▣ Raw Chronos-2 VaR 1% on the S&P 500: 1.67% breaches

Leads to

- ▣ Text as data (Chapter 15)

Chapter 15: LLMs and Sentiment Analysis

Key idea

- From dictionaries to FinBERT and large language models (LLMs): text becomes a number
- Text embeddings enter forecasting models as features

Key formula

- $S_t = \frac{N_t^+ - N_t^-}{N_t^+ + N_t^-}$, with $N_t^\pm$ the number of positive and negative texts on day t
- $r_{t+1} = a + b S_t + \varepsilon_{t+1}$ (predictive regression)

Questions the chapter answers

- Does a sentiment score predict returns or volatility out of sample, after costs?
- Was the text available at time t , or does it leak future information?

Leads to

- Sentiment in crypto markets and bubbles (Chapters 16–17)

Chapter 16: Digital Assets and DeFi

Key idea

- ▣ Cryptocurrencies are a statistically distinct asset class, mainly through their heavy tails [Pele et al. \(2023\)](#)
- ▣ Stablecoins, automated market makers and spot ETFs connect crypto to mainstream finance; decentralised finance (DeFi) runs on smart contracts

Key formula

- ▣ $x \cdot y = k$ (constant-product market maker; x, y : the two token reserves)
- ▣ $\text{Depeg} = |P_t - 1|$, with P_t the stablecoin price in USD

Questions the chapter answers

- ▣ How often and how far do stablecoins leave their peg?
- ▣ Did spot ETFs change the risk and the correlations of Bitcoin?

Leads to

- ▣ Explosive price episodes (Chapter 17)

Chapter 17: Bubbles and Crashes

Key idea

- ▣ A bubble is a price path that grows faster than fundamentals can justify
- ▣ Right-tailed unit-root tests date the start and the end of explosive episodes

Key formula

- ▣ $GSADF = \sup_{r_2 \in [r_0, 1]} \sup_{r_1 \in [0, r_2 - r_0]} ADF_{r_1}^{r_2}$
- ▣ $ADF_{r_1}^{r_2}$: the ADF statistic on the sample fraction $[r_1, r_2]$; r_0 : minimum window

Questions the chapter answers

- ▣ Which crypto or technology price episodes were explosive, and when did they end?
- ▣ Can a log-periodic model predict the crash date, or only fit it afterwards?

Leads to

- ▣ Contagion from one market to the system (Chapter 18)

Chapter 18: Systemic Risk and Networks

Key idea

- Systemic risk: losses of one institution that spread to the whole system
- CoVaR (the VaR of the system conditional on the distress of one institution), SRISK and spillover networks measure the contribution of each institution

Key formula

- $\Delta \text{CoVaR}^i = \text{CoVaR}_{\alpha}^{\text{sys}|X^i = -\text{VaR}_{\alpha}^i} - \text{CoVaR}_{\alpha}^{\text{sys}|X^i = \text{median}}$
- CoVaR: the VaR of the system return, conditional on the state of institution i

Questions the chapter answers

- Which banks contribute most to the tail risk of the system, and does it change in crises?
- How connected are Romanian and euro-area markets in stress?

Leads to

- Stress testing and the team projects

Recap: Learning, Text and New Markets

- ▣ Flexible models need a strict process: purging, embargo, deflated performance measures
- ▣ Foundation models tie with strong benchmarks on volatility and do not replace GARCH-based FHS for tail risk
- ▣ Text, crypto, bubbles and systemic risk reuse the same statistical toolkit
- ▣ New data and new models: the same five questions apply

Six Pitfalls That Returned in Every Chapter

Pitfalls of design

- ▣ **Look-ahead bias:** information from after t used at t
- ▣ **Overfitting:** a model tuned to the sample it is judged on
- ▣ **Multiple testing:** the best of many tries reported as if it were the only one

Pitfalls of data and inference

- ▣ **Wrong calendar join:** returns computed before aligning assets
- ▣ **Heavy tails ignored:** standard errors that assume the Normal distribution and independence
- ▣ **Wrong loss:** a loss function that cannot rank the forecast target

Pitfall 1: Look-Ahead Bias

- Any parameter, threshold or scaling estimated on the full sample leaks the future
 - ▶ Examples: full-sample GARCH parameters, full-sample standardisation, labels that overlap the test set
- BET, VaR 1%, 2005–2026: GARCH- t with full-sample parameters breaches on 1.31% of days, with rolling estimation on 1.47%
 - ▶ The look-ahead version looks better than any forecaster could have done in real time
- Chapter 13: shuffled K-fold reports 69.4% accuracy on pure noise
- Remedy: rolling or expanding windows; purging and embargo; write the information set $\mathcal{F}_{t-1}$ next to every forecast

Pitfall 2: Overfitting

- The in-sample fit rewards noise that will not repeat
 - ▶ Chapter 4: in-sample Sharpe 1.57, true Sharpe 0.30 for the same portfolios
- Backtest overfitting: trying many strategies and keeping the best [Bailey et al. \(2017\)](#)
 - ▶ The deflated Sharpe ratio corrects for the number of trials and for non-normal returns [Bailey & López de Prado \(2014\)](#)
- Remedy: fix the design before looking at the test period; report every model you tried; compare with a simple benchmark (1/N, historical simulation, random walk, log-HAR)

Pitfall 3: Multiple Testing

- Chapter 3: with 300 useless factors, on average 15.0 pass $|t| > 1.96$, and the best reaches $|t| \approx 3.06$
 - ▶ This is why the factor literature asks for $|t| > 3$ [Harvey, Liu & Zhu \(2016\)](#)
- Calendar anomalies, trading rules, model specifications: each extra test adds a chance of a false discovery
- Remedy: count your tests and correct
 - ▶ Bonferroni and Holm control the chance of any false discovery; Benjamini–Hochberg controls their share
 - ▶ For forecasts: the model confidence set instead of many pairwise tests

Pitfall 4: Joining on Returns Instead of Prices

- Bitcoin trades every day, SPY only on weekdays
 - ▶ Wrong: compute each return on its own calendar, then keep the common days: Bitcoin's weekend moves disappear
 - ▶ Right: keep the common days of the **prices**, then compute returns: Monday covers Friday to Monday for both
- Bitcoin, 2020–2026, on the common days: cumulative return 1025% (right) against 680% (wrong)
 - ▶ Monday standard deviation 5.04% (right) against 3.83% (wrong): the wrong join hides weekend risk
- Remedy: align prices on common days first; one series alone stays on its own calendar

Pitfall 5: Heavy Tails Ignored in Inference

- ▣ Classical standard errors assume independence and finite fourth moments
 - ▶ BET excess kurtosis 11.1, with a block-bootstrap 95% CI of [8.0, 13.9]: wide even with 6 670 days
- ▣ BET GARCH persistence: 95% CI [0.974, 1.007], so the half-life lies between 26 days and infinity
 - ▶ A point estimate of the half-life says little on its own
- ▣ Remedy: robust (HAC) errors, block bootstrap, tests designed for heavy tails; report intervals, not only points

Pitfall 6: A Loss Function That Cannot Rank the Target

- Variance forecasts are compared with a noisy proxy (r_t^2 or RV_t)
 - ▶ MSE and QLIKE rank them consistently; QLIKE is more robust to extreme days
 - ▶ Chapter 9: log-HAR beats GARCH in QLIKE; the MSE comparison is inconclusive
- ES alone cannot be ranked by any scoring function; the pair (VaR, ES) can [Fissler & Ziegel \(2016\)](#)
 - ▶ Use FZ0 with Diebold–Mariano tests [Diebold & Mariano \(1995\)](#); expect finite-sample limits on precision [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)
- Remedy: choose the loss before the comparison, and state it in the project report

Recap: A Checklist Against the Pitfalls

Design

- ▣ Every forecast uses only $\mathcal{F}_{t-1}$
- ▣ Test period fixed in advance and used once
- ▣ All models tried are reported; tests are counted

Data and inference

- ▣ Prices joined before returns; adjustments and calendars documented
- ▣ Robust or bootstrap uncertainty for every headline number
- ▣ Loss function consistent for the target; comparison with a simple benchmark

One Question, Four Chapters

The question

- How much can a BET index position lose tomorrow, with probability 1%?
- Which forecast of that loss would we have trusted in real time, 2005–2026?

Data

- BET index, official daily closing values, 6 Jan 2000 – 18 Sep 2026: 6670 daily log returns
- Days with an unchanged close (holidays filled with the previous value) removed

The four steps

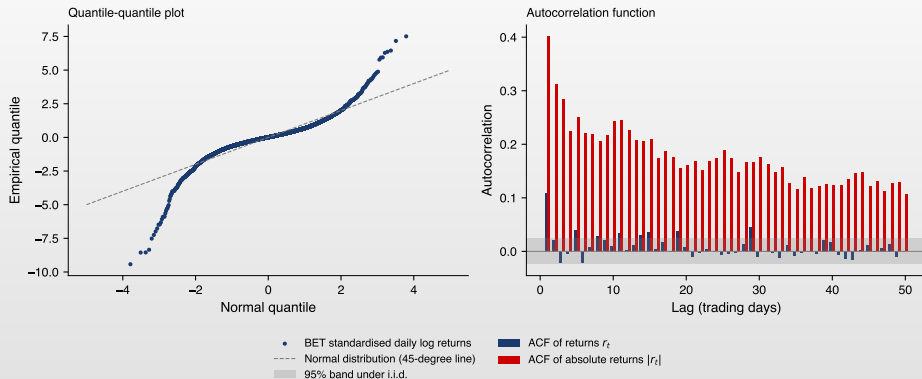
- Chapter 1: stylised facts, with robust inference
- Chapter 5: GARCH(1,1) with Student- t innovations
- Chapter 7: one-day VaR 1% from four models
- Chapter 8: coverage, independence and Basel zones



The Stock Exchange Palace, Bucharest (2023)

Photo: Chainwit. (2023); CC BY 4.0; Wikimedia Commons

Step 1: Stylised Facts of the BET

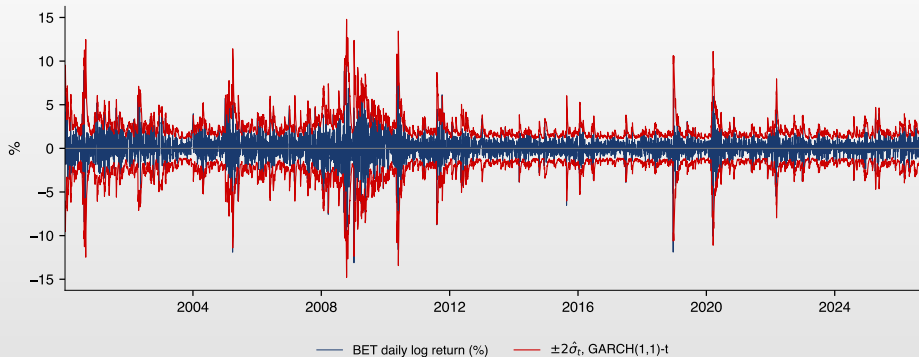


□ Left: quantiles of standardised returns against the Normal distribution; right: autocorrelations of r_t and $|r_t|$, lags 1–50

Reading Step 1

- ▣ Heavy tails: excess kurtosis 11.1 (block-bootstrap 95% CI [8.0, 13.9]); skewness -0.65
 - ▶ 51 days beyond ± 4 standard deviations, against 0.42 expected under the Normal distribution
 - ▶ Worst day: -13.1% (log) on 7 Jan 2009
- ▣ Clustering: first-order autocorrelation of $|r_t|$ is 0.40 and decays slowly; Ljung–Box $Q(10)$ on r_t^2 is 2440
 - ▶ Returns themselves have first-order autocorrelation 0.11: the BET is less liquid than large markets (Chapters 2, 10)
- ▣ Consequence: a Normal, constant-volatility VaR will be too low in stress and too high in calm periods

Step 2: GARCH(1,1)- t Volatility



- BET daily log returns with $\pm 2\hat{\sigma}_t$ from a GARCH(1,1) with Student- t innovations, estimated on 2000–2026

Reading Step 2

- ▣ Estimates (robust standard errors): $\hat{\alpha} = 0.192$ (0.025), $\hat{\beta} = 0.799$ (0.024), $\hat{\nu} = 5.01$
 - ▶ $\hat{\nu} \approx 5$: even after filtering, shocks have heavy tails
- ▣ Persistence $\hat{\alpha} + \hat{\beta} = 0.9907$: a half-life of about 74 trading days
 - ▶ But the 95% CI of the persistence includes 1 (Pitfall 5)
- ▣ Peak daily volatility 7.39% on 15 Oct 2008; at the end of the sample 1.67%

Step 3: Four VaR 1% Forecasts

Unconditional (last 500 days)

- ▣ **HS**: the loss $L = -r$ exceeded on 1% of the last 500 days (empirical quantile)
- ▣ **Normal**: $\text{VaR} = -(\hat{\mu} + \hat{\sigma} z_{0.01})$, $z_{0.01} = -2.326$

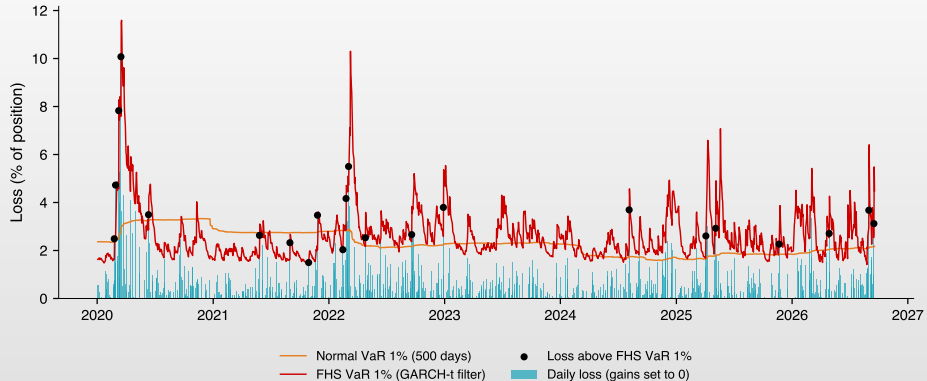
Conditional (GARCH- t on the last 1000 days)

- ▣ **GARCH- t** : $\text{VaR}_t = -(\hat{\mu} + \hat{\sigma}_t t_\nu^{-1}(0.01) \sqrt{(\nu - 2)/\nu})$
- ▣ **FHS**: $\text{VaR}_t = -(\hat{\mu} + \hat{\sigma}_t \hat{q}_{0.01}(z))$, the 1% quantile of the standardised residuals [McNeil & Frey \(2000\)](#)

Real-time design

- ▣ Forecast for day t uses data up to $t - 1$ only
- ▣ GARCH re-estimated every 21 trading days; $\hat{\sigma}_t$ updated daily
- ▣ Evaluation: 3 Jan 2005 – 18 Sep 2026, $T = 5\,432$ days
- ▣ Expected breaches at 1%: about 54

Step 3: Losses and VaR 1%, 2020–2026

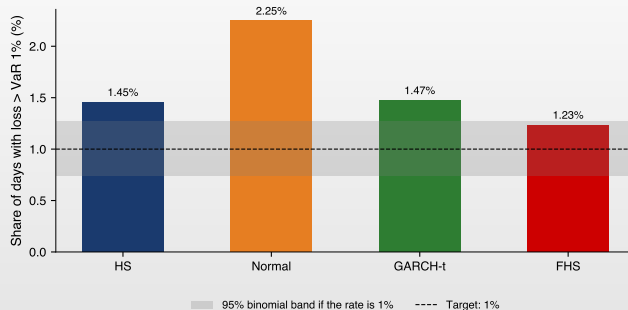


□ Daily BET losses (gains set to 0), Normal and FHS VaR 1%; black dots: losses above the FHS VaR

Reading Step 3

- FHS reacts within days: it peaked at 11.6% on 18 Mar 2020, during the COVID-19 crash
 - ▶ The Normal VaR on a 500-day window moves only slowly
- 2020–2026 (1 675 days, about 17 breaches expected): FHS 22 breaches, Normal 41
 - ▶ Normal breaches cluster in the first weeks of every stress episode
- Last forecast in the sample: FHS 4.48%, Normal 2.18%: the two models disagree most when volatility changes

Step 4: Breach Rates



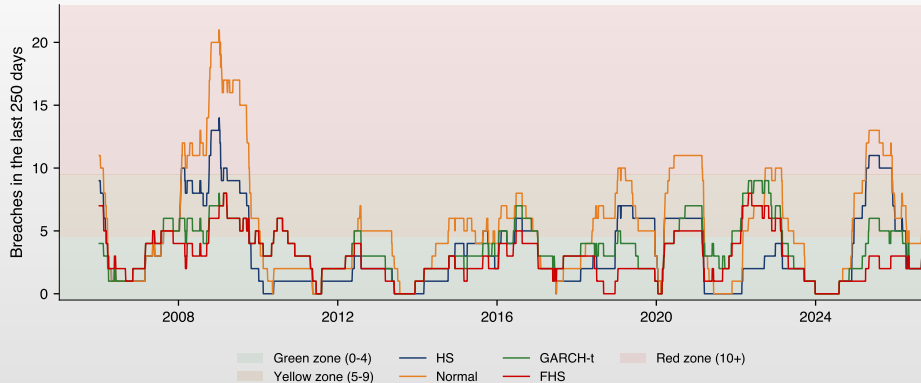
- Share of days with a loss above the VaR 1% forecast, 2005–2026; grey band: 95% binomial range if the true rate is 1%

Reading Step 4: Coverage and Independence

Model	Breaches	Rate	p_{uc}	p_{ind}	p_{cc}	Red zone
HS	79	1.45%	0.002	<0.001	<0.001	5.5%
Normal	122	2.25%	<0.001	<0.001	<0.001	18.6%
GARCH-t	80	1.47%	0.001	0.864	0.005	0.0%
FHS	67	1.23%	0.095	0.267	0.134	0.0%

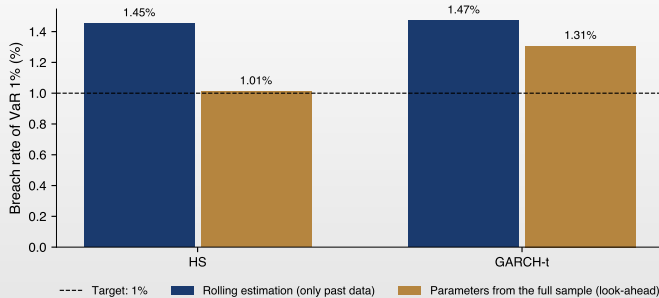
- ▣ p_{uc} : Kupiec coverage test; p_{ind} : Christoffersen independence test; p_{cc} : both jointly [Kupiec \(1995\)](#), [Christoffersen \(1998\)](#)
- ▣ Red zone: share of 250-day windows with 10 or more breaches (Basel traffic light)
- ▣ Only FHS passes all three tests at 5%; HS and Normal also fail independence: their breaches cluster
- ▣ GARCH- t has independent breaches but too many of them: the Student- t tail is not heavy enough at 1%

Step 4: Breaches in the Last 250 Days



- Number of breaches in a moving 250-day window, with the Basel green, yellow and red zones
- The Normal model spends 18.6% of the windows in the red zone, HS 5.5%, the two GARCH-filtered models never

The Look-Ahead Trap in the Case Study



- Full-sample HS quantile: 1.01% breaches, by construction; full-sample GARCH-t: 1.31%
- Both look better than their real-time versions: the gap is the value of information nobody had

Recap: The Case Study

- ▣ The BET shows every stylised fact: heavy tails, clustering, some linear autocorrelation
- ▣ GARCH- t captures clustering; the filtered residuals still have heavy tails
- ▣ A GARCH filter with an empirical tail (FHS) is the only VaR 1% that passes coverage and independence
- ▣ Real-time design matters: full-sample estimates flatter every model
- ▣ The same four steps are a template for your project

What the Project Is

- An empirical study of one question about financial markets, answered with the methods of this course
 - ▶ Teams; four steps: replicate, extend, deliver on GitHub, present and defend (next slide)
- It carries **70%** of the final grade; quizzes and seminar activity carry 20%, attendance 10%
- Starting points: the Part C question of every seminar (next section)
 - ▶ Your own question is welcome if it passes the five questions of this chapter
- A good project can end with a negative answer, if the evidence is honest and precise

The Project in Four Steps

- ▣ **1. Replicate:** reproduce one published number (a table, a coefficient, a backtest result) from a paper or from the lecture
 - ▶ An AI assistant cannot guess an exact published figure: matching it shows that your data and code are right
- ▣ **2. Extend:** new data, a new market (BVB, crypto) or a more recent sample
 - ▶ One clear question, answered with a test, not only with a chart
- ▣ **3. Deliver on GitHub:** code, data description and a README that reproduces every result
 - ▶ `AI_USE.md` and `AI_ERRORS.md` are mandatory
- ▣ **4. Present and defend:** team presentation, then a 10-minute oral defence
 - ▶ Each member explains one result and answers a “what changes if...” question; grades can differ between members

Choosing the Question

A good question is

- **Specific:** one market, one period, one quantity
- **Testable:** a null hypothesis and a benchmark
- **Feasible:** data you can obtain and clean in a few weeks
- **Relevant:** an answer matters to an investor, a bank or a regulator

Examples

- Too vague: "Is Bitcoin risky?"
 - ▶ Better: "Did spot ETFs change the one-day VaR 1% of Bitcoin, and which model forecasts it best since 2024?"
- Too broad: "Can machine learning predict the market?"
 - ▶ Better: "Does a boosting model beat log-HAR for next-day BET volatility, in QLIKE, 2018–2026?"

Data

- Use the course data where possible: daily series of indices, ETFs, BVB stocks, crypto and FX, plus intraday bars
 - ▶ Public sources: BNR reference rates, FRED, Kenneth French Data Library, DefiLlama; the BVB archive for Romanian indices
- Apply the course conventions
 - ▶ Adjusted prices for ETFs and stocks, closing values for indices, FX and crypto
 - ▶ Remove holiday fills and weekend quotes where the market is closed; annualise with the real frequency
 - ▶ Several assets: join on prices, then compute returns
- Look for bad prints before modelling: jumps of $\pm 50\%$ in adjusted prices are usually data errors

Methods: From Question to Chapter

If your question is about	Start from
Return distribution, tails, predictability Expected returns, factors, portfolios	Chapters 1–2: moments, robust tests, variance ratios Chapters 3–4: Fama–MacBeth, shrinkage, out-of-sample Sharpe tests
Volatility and correlation Risk measures and capital	Chapters 5, 6, 9: GARCH, DCC, copulas, log-HAR Chapters 7–8: FHS, EVT, backtests, FZ0
Liquidity, options, continuous time	Chapters 10–12: Amihud, Black–Scholes, VIX, variance risk premium
Forecasting with ML, deep learning or text Crypto, bubbles, systemic risk	Chapters 13–15: purged CV, foundation models, sentiment Chapters 16–18: stablecoins, GSADF, CoVaR, networks

Honest Evaluation Checklist

Before you look at the results

- ☐ Write the hypothesis, the benchmark and the loss function
- ☐ Fix the estimation window and the test period
- ☐ Decide how many models and parameters you will try

When you report

- ☐ Out-of-sample results only for claims about forecasting
- ☐ Every headline number with a standard error, CI or p -value
- ☐ Costs for trading strategies; subperiods and a second market for robustness
- ☐ What did **not** work, and why



The Report

- ▣ **1. Question and motivation:** one paragraph, one hypothesis
- ▣ **2. Literature:** three to eight clickable references, including the course textbooks [Franke, Härdle & Hafner \(2019\)](#), [McNeil, Frey & Embrechts \(2015\)](#)
- ▣ **3. Data and replication:** sources, period, cleaning rules, descriptive statistics; the published number and your value
- ▣ **4. Methods:** models, estimation window, benchmark, loss function, tests
- ▣ **5. Results:** tables and charts, each with an interpretation
- ▣ **6. Robustness and limitations; 7. Conclusion:** the answer, in one sentence
- ▣ **Code:** a GitHub repository whose README and notebook reproduce every number, with [AI_USE.md](#) and [AI_ERRORS.md](#)

The Presentation

Structure (about 12 minutes)

- ▣ The question and why it matters (1 slide)
- ▣ Data and one chart that shows the problem (2 slides)
- ▣ Method and benchmark (2 slides)
- ▣ Main result with its uncertainty (2–3 slides)
- ▣ Robustness, limitations, answer (2 slides)

Good practice

- ▣ One message per slide; charts with readable axes and the legend below
- ▣ Every team member presents and can answer questions on the code
- ▣ Then a 10-minute oral defence: each member explains one result and answers a “what changes if...” question, without AI

Project Grading (70% of the Final Grade)

Component	What we look for
Replication (step 1)	A published number reproduced; any gap explained
Question, data and extension (step 2)	Specific, testable question; documented data and calendar alignment
Methods and inference	Real-time design, a benchmark, out-of-sample tests, uncertainty
Interpretation	Economic meaning, limitations, honest conclusion
GitHub repository (step 3)	README that reproduces every result; AI_USE.md, AI_ERRORS.md
Presentation and oral defence (step 4)	Individual: one result explained, answers to “what changes if...”
Total	70% of the final grade; plus quizzes and activity (20%) and attendance (10%)

- Grades can differ between team members, depending on the oral defence

Timeline

- Week 1 after this chapter: team and question
 - ▶ Submit a one-page proposal (template in Seminar 19, Part C); open the GitHub repository
- Weeks 2–3: data and the replication of one published number (step 1)
- Weeks 4–5: extension, evaluation, robustness (step 2); feedback at the seminar
- Final weeks: repository with README and the two AI files (step 3); presentation and oral defence (step 4)
- Exact dates are announced at the course and seminar

Where Projects Usually Lose Points

- ▣ A question that changes after the results are known
- ▣ In-sample fit presented as forecasting power
- ▣ No benchmark, or a weak one chosen to be beaten
- ▣ Numbers without uncertainty; many tests without correction
- ▣ Charts without units, legends inside the data, unexplained acronyms
- ▣ Code that does not run, or numbers in the report that the code does not produce
- ▣ AI output kept without checks: undeclared use, unchecked code, references that do not exist

Recap: The Team Project

- ▣ One specific, testable question; one benchmark; one loss function chosen in advance
- ▣ Clean data with the course conventions; real-time design; honest reporting
- ▣ Replicate, extend, deliver on GitHub, present and defend: 70% of the grade

The AI Policy of the Course

- ▣ **Allowed:** AI assistants for code, debugging, literature search and writing, in the project and in the seminars
- ▣ **Declared:** every tool, what it was used for and what you checked yourselves, in [AI_USE.md](#)
 - ▶ Undeclared use of AI counts as plagiarism
- ▣ **Checked:** you are responsible for every line of code and text
 - ▶ “The AI wrote it” is not an explanation at the oral defence
 - ▶ Every reference needs a working DOI or link; a reference that does not exist counts as fabricated data
- ▣ **No AI during the oral defence:** you answer alone, from your own understanding of the project

The Two Mandatory Files

AI_USE.md: the declaration

- ▣ Tools: name, version, who used them and for what
- ▣ Files or sections where AI-written code or text was kept, and what you changed
- ▣ How you verified the replicated number, the data, the formulas and the references
- ▣ What the team did without AI

▣ Templates in English and Romanian: github.com/danpele/MFM/tree/main/project

AI_ERRORS.md: the error log

- ▣ At least **three** places where the assistant was wrong
- ▣ For each: what you asked, what it answered, what was wrong, how you found out, the correct version
- ▣ Errors you caught are a strength of the project, not a weakness

Typical AI Errors in Financial Modelling (1/2)

- ▣ **Look-ahead bias:** a rolling statistic or a scaler that includes day t in the forecast for day t
 - ▶ Check: change the last observation; the forecast for that day must not move; shift by one day (`.shift(1)`)
- ▣ **Sign of VaR:** the 1% quantile of returns reported as VaR, or the 99% quantile of returns (a gain) used as the loss
 - ▶ Check: $\text{VaR}_\alpha = -q_\alpha(r) > 0$ for a loss; VaR 1% of a daily index is a few percent, not negative
- ▣ **Annualisation:** daily volatility multiplied by 252 instead of $\sqrt{252}$, or 252 days used for crypto that trades every day
 - ▶ Check: means scale with h , volatilities with $\sqrt{h}$; use the real number of trading days (365 for crypto)

Typical AI Errors in Financial Modelling (2/2)

- ❑ **Random cross-validation (CV) on time series:** shuffled K-fold trains on the future; in Chapter 13 it reports 69.4% accuracy on pure noise
 - ▶ Check: is any test date earlier than a training date? Use walk-forward or purged CV with an embargo [López de Prado \(2018\)](#)
- ❑ **Invented references:** plausible authors, title and journal, but the paper does not exist
 - ▶ Check: search every DOI on [Crossref](#); open the paper and find the claim you cite
- ❑ **Also frequent:** wrong degrees of freedom in a test, library defaults you did not choose, numbers stated without running any code
 - ▶ Every seminar has a C2 exercise that trains you to catch such errors

Recap: Using AI in Your Project

- Use AI freely, declare it in `AI_USE.md`, and check every line it writes
- The replicated number is your best test: an AI cannot guess it
- Log at least three AI errors in `AI_ERRORS.md`; at the oral defence you answer alone

Project Seeds: Part C of Every Seminar (1/2)

Ch.	Open question (link to the seminar)
0	Did spot ETFs change Bitcoin's correlation with equities?
1	Did BET-TR autocorrelation fall after the FTSE reclassification?
2	Did the BVB become more efficient after the 2020 FTSE upgrade?
3	Do factor premia exist on the Bucharest Stock Exchange?
4	An optimal portfolio of BVB blue chips in RON
5	Is the Romanian market's volatility driven by the US?
6	Did Romanian and euro-area equities become more integrated after 2020?
7	Capital for a BVB portfolio vs an S&P 500 portfolio

- Each seminar gives the data, a starting method and a reference analysis discussed in class

Project Seeds: Part C of Every Seminar (2/2)

Ch.	Open question (link to the seminar)
8	Which risk model would you certify for a Romanian bank's trading book?
9	Is Bitcoin's volatility more predictable than the S&P 500's?
10	Does illiquidity carry a premium on the BVB?
11	Which continuous-time model fits BET and Bitcoin?
12	Is the variance risk premium a tradable signal?
13	Does a foundation model beat boosting?
14	Can a zero-shot foundation model replace GARCH for the BET?

- ▣ Seminars 15–18 add questions on sentiment, digital assets, bubbles and systemic risk

Turning a Seed into a Project

- Replicate first: one published number from a paper behind the seed (step 1)
- Narrow it: one market, one period, one number to forecast or test
 - ▶ Example: Seminar 8 becomes “Which of HS, FHS and GARCH-EVT gives a correct VaR 1% for a 50/50 BET–EUR portfolio in 2015–2026?”
- Extend it: a second market, a new model, a longer or more recent sample
 - ▶ Example: add a foundation model (Chapter 14) as a volatility filter inside FHS
- Keep the reference analysis as your benchmark and try to improve on it honestly

An Open Question

How much of a result is the data, and how much the researcher?

- 164 teams tested the same hypotheses on the same data: the dispersion of their answers, the **nonstandard errors**, was sizable and underestimated by the teams [Menkveld et al. \(2024\)](#)
 - ▶ It was smaller for more reproducible research and fell after peer-review stages
- Replication is not hopeless: in a Bayesian model of replication most asset-pricing factors replicate and work out of sample in 93 countries [Jensen, Kelly & Pedersen \(2023\)](#)
- AI assistants make every extra specification almost free: the forking paths multiply, but so does the ability to report all of them
 - ▶ Formal version for a project: the conclusion must hold over a pre-registered set S of defensible specifications, with the family corrected (FWER or FDR); report its distribution over S , not the best cell

The Discovery Loop, Step by Step (1/2)

- **Week 1, the question:** sharpen a seed into H_0/H_1 with a replication target
 - ▶ “Here is our seed from Part C of Seminar N. Give three sharper, testable versions, the published number each would replicate and the series in data/market it needs.”
- **Weeks 1–2, literature and target:** the number to replicate is one you have seen in the PDF
 - ▶ “List the most cited and the most recent papers on this question with DOI; for each, the headline estimate, its table and its sample.”
 - ▶ Resolve every DOI on [Crossref](#); search with Semantic Scholar, Elicit or Google Scholar; open the table yourselves
- **Week 2, hypotheses and pre-registration,** committed before the first result
 - ▶ “For our H_0 and H_1 , list competing hypotheses and the implication that separates each; then draft a pre-registration: sample, filters, windows, tests, correction.”
 - ▶ The commit history of the repository is the evidence that the plan came first

The Discovery Loop, Step by Step (2/2)

- **Weeks 2–3, code and replication:** the replicated number becomes a test in the repository
 - ▶ `“Write a function that reproduces the published number from data/market and an assertion with the tolerance we accept.”`
- **Weeks 4–5, extension and robustness:** run the whole pre-registered grid
 - ▶ `“Here are all the cells we ran. Which conclusions survive a Holm correction, and which ones would a referee call data snooping?”`
- **Before the defence, critique:** the assistant asks, the team answers
 - ▶ `“Act as the examiner of our oral defence: ask the ten hardest questions on identification, inference and data. Do not answer them.”`
 - ▶ The team answers without AI, as at the defence itself
- **Report:** the distribution of results over the grid, not the best cell; prompts in `AI_USE.md`, caught errors in `AI_ERRORS.md`

What the Human Must Check

- **The replication target exists:** number, table and sample taken from the paper, not from the assistant's summary
 - ▶ Test: the report cites the page and the table; a second member finds the same number there
- **The plan came first**
 - ▶ Test: in the commit history, the pre-registration precedes the first results file
- **Forking paths:** more specifications tried than reported
 - ▶ Test: every run writes its configuration and result to a log; the report states their number and corrects the family
- **Reproducibility:** numbers typed into the report by hand
 - ▶ Test: a fresh clone and one command regenerate every number to the reported digit
- **Understanding:** each member can derive the key formula and defend every choice
 - ▶ Test: an internal mock defence, without AI, one week before the real one

Mini-Case: Replicate, Then Extend

- **Replicate:** FHS VaR 1% for the BET; Chapter 8 reports 59 breaches in 4 937 days (1.20%, Kupiec $p = 0.181$)
 - ▶ The case study of this chapter: 67 in 5 432 days (1.23%, $p = 0.095$); same model, different evaluation start (2007 vs 2005) and settings: document every gap before calling a replication failed
- A typical assistant proposal, reading each p -value alone at 5%: “only GARCH-EVT is correctly calibrated on all four assets”
- **Extend:** the 24 conditional-coverage tests of Chapter 8 (4 assets $\times$ 6 models) form one family
 - ▶ Rejections at 5%: 12 without correction, 10 with Holm (FWER), 11 with BH (FDR 5%)
 - ▶ FHS fails only on the S&P 500 ($p = 0.040$); within the family of 24 it survives, so after Holm FHS and GARCH-EVT pass on all four assets
- Interpretation: “FHS fails for the S&P 500” depends on whether the test is read alone or in its family; pre-register the family and report both

Your Turn: the Capstone Plan

- **Question:** your seed from Part C, written as H_0/H_1 , with the observation that would falsify it
- **Replicate first:** one published number seen in the paper, turned into a passing test in the repository
 - ▶ Then extend over a pre-registered grid, with the family corrected and the full distribution reported
- **Deliverables** along the timeline: proposal and replication target, pre-registration commit, passing replication test, grid of results, mock defence, report, oral defence
- **AI log:** the prompts that changed the design in AI_USE.md; every caught error, with the test that caught it, in AI_ERRORS.md ([templates](#))

Key Takeaways

- ▣ Returns are hard to predict; their risk is not: volatility, tails and dependence can be forecast
- ▣ Every model is judged out of sample, against a simple benchmark, with a loss that fits the target
- ▣ Heavy tails change inference: use robust errors and the bootstrap, report intervals
- ▣ Most wrong results come from the design, not from the model: look-ahead, overfitting, multiple testing, calendars
- ▣ New tools (ML, foundation models, LLMs) raise the bar for evaluation, not lower it

Further Study

Textbooks

- Franke, Härdle & Hafner (2019); Härdle, Chen & Overbeck (2017); McNeil, Frey & Embrechts (2015); Tsay (2010)

Advanced courses on quantinar.com

-  Statistics of Financial Markets
-  Measuring Statistical Risk

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_19

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